

Mathematics for Social Scientists

Part 2: Introduction to Analysis Sequences, Limits, Convergence

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Outline

Limits

Continuous Functions

Sequences, Convergence and Series

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Limits

Continuous Functions

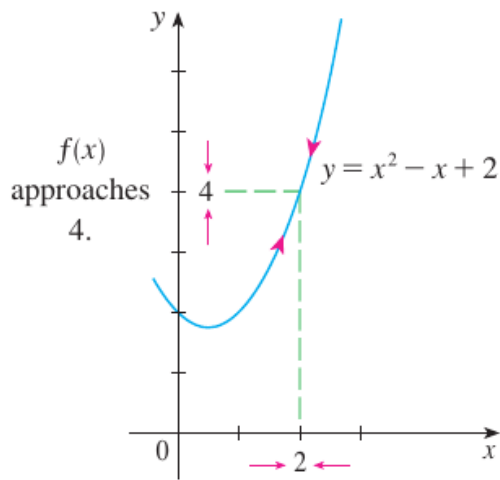
Sequences, Convergence and Series

The limit of a function

Consider $f(x) = x^2 - x + 2$ for x near 2 (but $x \neq 2$).

x	$f(x)$
1.90	3.710000
1.95	3.852500
1.99	3.970100
1.995	3.985025
1.999	3.997001

x	$f(x)$
2.10	4.310000
2.05	4.152500
2.01	4.030100
2.005	4.015025
2.001	4.003001



As x approaches 2,
As $x \rightarrow 2$, $f(x) \rightarrow 4$.

Notation and informal definition

Definition 1 (Limit of a function).

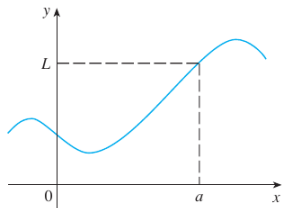
Suppose $f(x)$ is defined when x is near the number a (on some open interval that contains a , except possibly at a). We write

$$\lim_{x \rightarrow a} f(x) = L$$

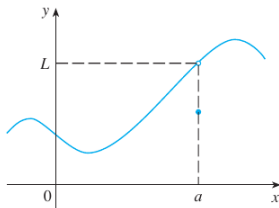
and say “*the limit of $f(x)$, as x approaches a , equals L* ” if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a (on either side of a) but not equal to a .

Alternative notation: $f(x) \rightarrow L$ as $x \rightarrow a$. (Only the behavior near a matters; $f(a)$ may be undefined.)

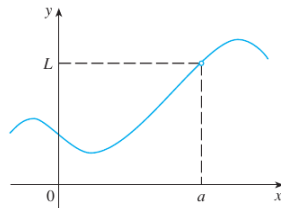
Behavior near a vs. value at a



(a)



(b)



(c)

(a) $f(a) = L$ (b) $f(a) \neq L$ (c) $f(a)$ undefined; in all three, $\lim_{x \rightarrow a} f(x) = L$.

Example

Guess $\lim_{x \rightarrow 1} \frac{x-1}{x^2-1}$.

$x < 1$	$f(x)$
0.50	0.666667
0.90	0.526316
0.99	0.502513
0.999	0.500250
0.9999	0.500025

$x > 1$	$f(x)$
1.50	0.400000
1.10	0.476190
1.01	0.497512
1.001	0.499750
1.0001	0.499975

For $x \neq 1$,

$$\frac{x-1}{x^2-1} = \frac{x-1}{(x-1)(x+1)} = \frac{1}{x+1} \Rightarrow \lim_{x \rightarrow 1} \frac{x-1}{x^2-1} = \frac{1}{2}.$$

Redefining the value at the point

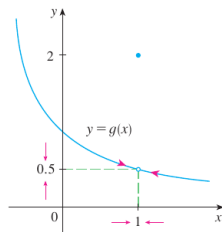
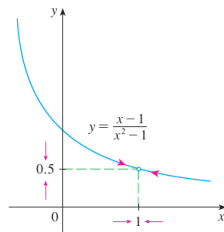
Define

$$g(x) = \begin{cases} \frac{x-1}{x^2-1}, & x \neq 1, \\ 2, & x = 1. \end{cases}$$

For $x \neq 1$,

$$g(x) = \frac{1}{x+1} \implies \lim_{x \rightarrow 1} g(x) = \frac{1}{2}.$$

Changing the single value $g(1)$ does not affect the limit.



One-sided limits (Informal)

Definition 2 (Left-hand limit).

We write $\lim_{x \rightarrow a^-} f(x) = L$ and say “the left-hand limit of $f(x)$ as x approaches a equals L ” if we can make $f(x)$ arbitrarily close to L by taking x sufficiently close to a with $x < a$.

Definition 3 (Right-hand limit).

We write $\lim_{x \rightarrow a^+} f(x) = L$ and say “the right-hand limit of $f(x)$ as x approaches a equals L ” if we can make $f(x)$ arbitrarily close to L by taking x sufficiently close to a with $x > a$.

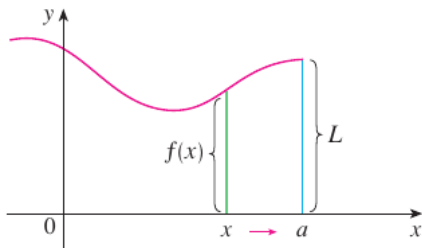
Notation: $x \rightarrow a^-$ means $x < a$; $x \rightarrow a^+$ means $x > a$.

One-sided and two-sided limits

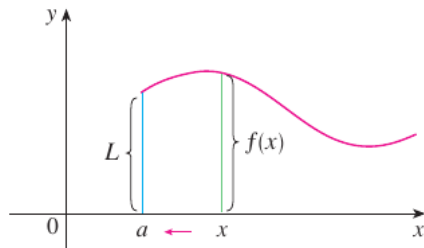
From our Definition 1 (Limit of a function):

Theorem 1.

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$



(a) $\lim_{x \rightarrow a^-} f(x) = L$



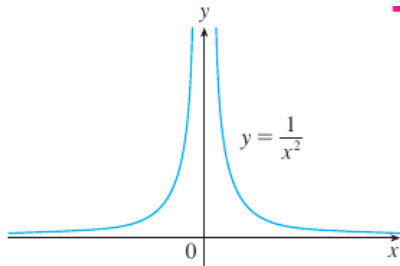
(b) $\lim_{x \rightarrow a^+} f(x) = L$

Infinite limits

Find $\lim_{x \rightarrow 0} \frac{1}{x^2}$ (if it exists).

As $x \rightarrow 0$, $x^2 \rightarrow 0$ and $\frac{1}{x^2}$ becomes arbitrarily large. The values of $f(x) = 1/x^2$ do not approach a finite number, so the limit does not exist as a real number. We indicate this behavior by writing

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$



Infinite limits (informal)

Definition 4 (Infinite limit).

Let f be a function defined on both sides of a , except possibly at a .

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a , but not equal to a .

Definition 5 (Negative infinite limit).

Let f be a function defined on both sides of a , except possibly at a .

$$\lim_{x \rightarrow a} f(x) = -\infty$$

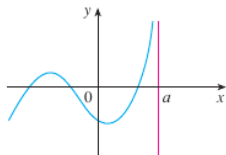
means that the values of $f(x)$ can be made arbitrarily large in magnitude and negative by taking x sufficiently close to a , but not equal to a .

One-sided infinite limits and vertical asymptotes

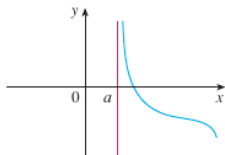
Definition 6 (Vertical asymptote).

The line $x = a$ is a vertical asymptote of $y = f(x)$ if at least one of the following holds:

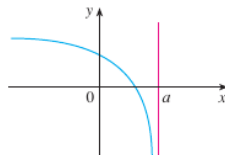
$$\lim_{x \rightarrow a^-} f(x) = \infty, \quad \lim_{x \rightarrow a} f(x) = -\infty, \quad \lim_{x \rightarrow a^-} f(x) = \pm\infty, \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty.$$



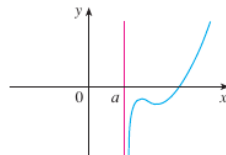
(a) $\lim_{x \rightarrow a^-} f(x) = \infty$



(b) $\lim_{x \rightarrow a^+} f(x) = \infty$



(c) $\lim_{x \rightarrow a^-} f(x) = -\infty$



(d) $\lim_{x \rightarrow a^+} f(x) = -\infty$

Properties of limits

Suppose c is a constant and the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist. Then:

1. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
2. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
3. $\lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$
4. $\lim_{x \rightarrow a} [f(x) g(x)] = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right)$
5. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$

Example: compute a limit by simplifying near the point

Example.

Find $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

Since $x^2 - 1 = (x - 1)(x + 1)$,

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1 \quad \text{for } x \neq 1.$$

Therefore

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2.$$

If $f(x) = g(x)$ for all x near a with $x \neq a$, then

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x),$$

provided the limits exist.

Precise ε - δ definition of a limit

Definition 7 (Limit at a point).

Let f be defined when x is near a (on some open interval containing a , possibly excluding a). We say

$$\lim_{x \rightarrow a} f(x) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Epsilon- and delta-neighborhoods on the graph

Fix $\varepsilon > 0$ and consider the ε -neighborhood of L :

$$(L - \varepsilon, L + \varepsilon).$$

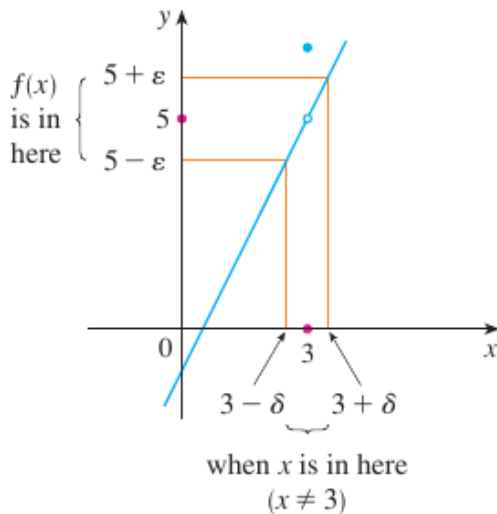
If $\lim_{x \rightarrow a} f(x) = L$, then there exists $\delta > 0$ such that
whenever

$$0 < |x - a| < \delta \quad \left(x \in (a - \delta, a + \delta) \setminus \{a\} \right),$$

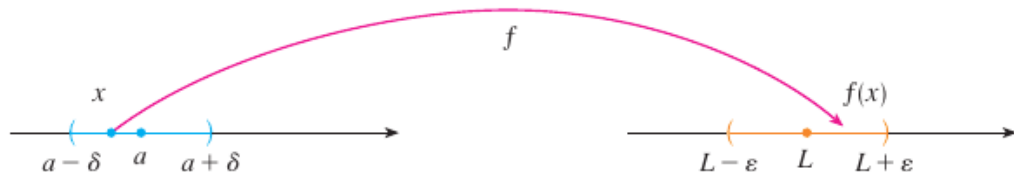
we have

$$f(x) \in (L - \varepsilon, L + \varepsilon).$$

If one δ works, any smaller δ also works.



Interval mapping interpretation



$$x \in (a - \delta, a + \delta) \setminus \{a\} \implies f(x) \in (L - \varepsilon, L + \varepsilon).$$

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Continuous functions

Some functions can be drawn without taking the pen from the paper.

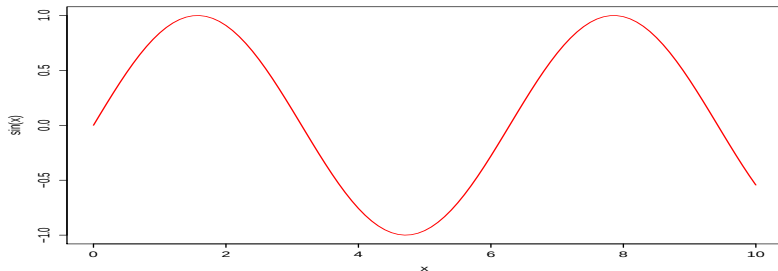


Figure:

This is the intuition behind *continuity*.

Definition of continuity I

From now on, let I be some interval (e.g., $I = (a, b)$, $I = [a, b)$).

Definition 8 (ε - δ definition of continuity).

A function $f : I \rightarrow \mathbb{R}$ is **continuous at** $a \in I$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|x - a| \leq \delta \quad \implies \quad |f(x) - f(a)| \leq \varepsilon$$

for all $x \in I$.

f is called **continuous** if it is continuous at a for all $a \in I$.

Definition of continuity II

Definition 9 (Sequence definition of continuity).

A function $f : I \rightarrow \mathbb{R}$ is **continuous at** $a \in I$ if for every sequence $(a_n)_{n \in \mathbb{N}}$ with limit a it holds that

$$\lim_{n \rightarrow \infty} f(a_n) = f(a).$$

Note: Continuity at a is a local property.

We can 'extend' the definition above to a universal property.

Definition 10.

A function $f : I \rightarrow \mathbb{R}$ is **continuous** if it is continuous at every point $x \in I$.

What continuity at a point requires

Definition of continuity at a :

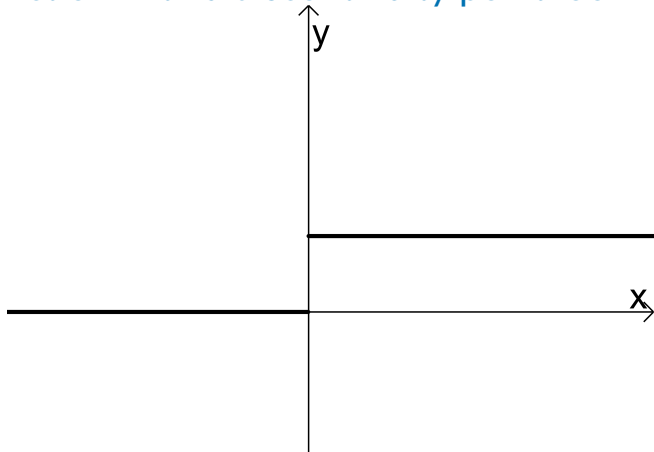
$$\lim_{x \rightarrow a} f(x) = f(a).$$

Implicitly, this requires three things:

1. $f(a)$ is defined (so a is in the domain of f),
2. $\lim_{x \rightarrow a} f(x)$ exists (finite),
3. The two agree: $\lim_{x \rightarrow a} f(x) = f(a)$.

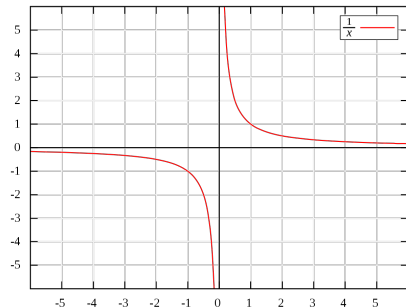
If any of these fail, f is *discontinuous at a* .

What does a function with a discontinuity point look like?

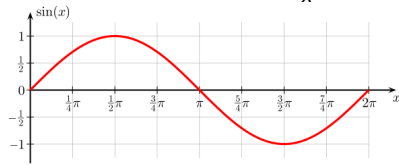


$$f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}$$

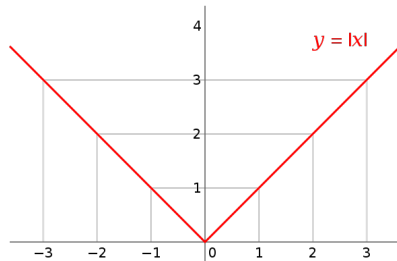
Examples of continuous functions



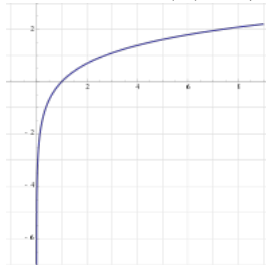
$$f: (0, \infty) \rightarrow \mathbb{R}, f(x) = \frac{1}{x}$$



$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sin(x)$$



$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x|$$



$$f: (0, \infty) \rightarrow \mathbb{R}, f(x) = \ln(x)$$

Examples of continuous functions

- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = a \in \mathbb{R}$
- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x$
- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$
- $f : (0, \infty) \rightarrow \mathbb{R}, f(x) = \frac{1}{x}$
- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x|$
- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sin(x)$
- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \exp(x)$
- $f : (0, \infty) \rightarrow \mathbb{R}, f(x) = \ln(x)$
- $f : [0, \infty) \rightarrow \mathbb{R}, f(x) = \sqrt{x}$

Combination and composition of continuous functions

Proposition 1.

Let $f, g : I \rightarrow \mathbb{R}$ be functions that are continuous at $a \in I$.

1. The functions

$$f + g, \quad f \times g$$

are continuous at a .

2. If $f(x) \neq 0$ for all $x \in I$, then the function $1/f$ is continuous at a .

Proposition 2.

Let $f : I \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be functions. Let $a \in I$.

If f is continuous at a and g is continuous at $b := f(a)$, then

$$g \circ f : I \rightarrow \mathbb{R} : x \mapsto g(f(x))$$

is continuous in a .

“A continuous function of a continuous function is a continuous function.”

Inverse of continuous functions

Theorem 2.

Let $f : I \rightarrow \mathbb{R}$ be a continuous and bijective function. Then the inverse function

$$f^{-1} : \mathbb{R} \rightarrow I : f(x) \mapsto x$$

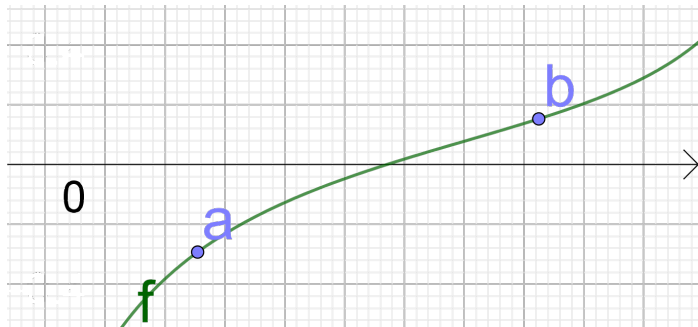
is also continuous.

Intermediate Value Theorem

Theorem 3.

Let $f : I \rightarrow \mathbb{R}$ be continuous on the interval I . Suppose $a, b \in I$ with $a < b$. If N is any number between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that

$$f(c) = N.$$



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Sequences

Recall: $\mathbb{N} = \{1, 2, 3, \dots\}$.

Definition 11.

A **sequence** of real numbers is a rule which assigns to each element $n \in \mathbb{N}$ a real number $a_n \in \mathbb{R}$. Notation: $(a_n)_{n \in \mathbb{N}}$ or $(a_n)_{n \geq 0}$ or simply (a_n) .

Note that a sequence can be interpreted as a function from $\mathbb{N}$ to $\mathbb{R}$.

Examples:

- (Constant sequence) $a_n = 1$ for all n
- $a_n = \frac{1}{n}$; $a_n = (-1)^n$; $a_n = n$
- (Exponential sequence) $a_n = (1 + \frac{1}{n})^n$
- (Fibonacci sequence) $a_1 = 1, a_2 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$

Convergent sequences

Definition 12.

A sequence $(a_n)_{n \in \mathbb{N}}$ **converges** if there is some $a \in \mathbb{R}$ such that *for all* $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_n - a| < \epsilon \quad \text{for } n \geq N.$$

In this case, a is called the **limit** of the sequence.

Notation: $a_n \rightarrow a$ or $\lim_{n \rightarrow \infty} a_n = a$

Convergent Sequences II

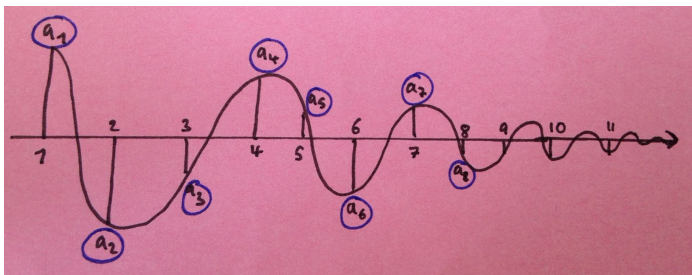


Figure: $a_n = \frac{\sin(n)}{n}$ or something similar.

Example.

- The sequence $a_n = \frac{1}{n}$ has limit 0.
- The sequence $a_n = (-1)^n$ has no limit.
- The sequence $a_n = n$ has limit $+\infty$. Such a sequence is called **divergent**.

Algebra of Limits

Proposition 3 (Algebra of limits).

Let $(a_n), (b_n)$ be convergent sequences with limits $a = \lim_{n \rightarrow \infty} a_n, b = \lim_{n \rightarrow \infty} b_n$. Let $C \in \mathbb{R}$ be a constant. Then the sequences $(a_n + b_n), (Ca_n), (a_nb_n)$ and $(|a_n|)$ converge and the limits are given by:

- $\lim_{n \rightarrow \infty} (a_n + b_n) = a + b,$
- $\lim_{n \rightarrow \infty} (Ca_n) = Ca,$
- $\lim_{n \rightarrow \infty} (a_nb_n) = ab,$
- $\lim_{n \rightarrow \infty} |a_n| = |a|.$

Remark: The algebra of limits above for sequences also works for functions:

If $\lim_{x \rightarrow a} f(x) = f, \lim_{x \rightarrow a} g(x) = g$, then:

- $\lim_{x \rightarrow a} (f(x) + g(x)) = f + g,$
- $\lim_{x \rightarrow a} (C.f(x)) = C.f,$
- $\lim_{x \rightarrow a} (f(x).g(x)) = f.g$, etc.

Series

Definition 13.

A **finite series** is the cumulative sum of a sequence, up to the k^{th} element for some $k < \infty$. The notation we use is

$$S_k = \sum_{n=1}^k a_n = a_1 + a_2 + \cdots + a_k.$$

An **infinite series** is

$$S_m = \sum_{n=1}^{\infty} a_n = \lim_{k \rightarrow \infty} S_k.$$

Series II

Series can also converge, but the definition is much simpler. A finite series **converges** if $\lim_{k \rightarrow \infty} S_k < \infty$.

Example.

It's not always easy to see whether a series converges.

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots = \infty, \quad \text{but} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \cdots < \infty.$$

Power series

Definition 14.

A **power series** (about a) is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

where (c_n) are real coefficients.

Taylor series

Some functions can be represented as infinite series. These are called **Taylor series**.

Examples. For $x \in \mathbb{R}$:

- $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$
- $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$
- $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$

A function can be *approximated* by using only finitely many terms from its Taylor series.